Fathom: Cognitive Measurement Instruments for Transformer Internals via SAE Feature Coherence Geometry

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Abstract

We present Fathom, a suite of cognitive measurement instruments for transformer internals built on sparse autoencoder (SAE) feature coherence geometry. Our contributions span measurement, diagnosis, and intervention:

Measurement. We introduce commitment intensity (S), derived as the inverse participation ratio (IPR) of per-token coherence events. S_{early} (first 7 tokens) discriminates hallucination from correct generation with pooled $d = 0.494$ across three independent designs (Fisher $p = 0.0008$), outperforms logit entropy head-to-head (AUC 0.663 vs. 0.607), and replicates across two architecture families (Fisher $p = 0.0006$). Three SAE-free probes across four architectures all fail (Fisher $p > 0.6$).

Diagnosis. We discover that the *shape* of per-layer coherence distributions carries discriminative information invisible to scalar summaries. The Gini coefficient of coherence across layers achieves AUC $= 0.685$ ($n = 200$, 13/16 metrics significant after Holm–Bonferroni correction), while scalar depth (K) is at chance. Cognitive autopsy localizes the fault: layer 11 is the primary locus in 63% of hallucinations ($p = 2.7 \times 10^{-12}$), with layers 11–12 accounting for 41% of all excess coherence. Hallucination is not a depth shift but a coherence *concentration*.

Intervention. Best-of-N sampling scored by a Cognitive Health Score (Gini + spectral spread + early-layer mass) improves on 100% of test prompts versus greedy decoding. Coordinated suppression of 220 hallucination-associated features across 22 layers produces 33% first-token changes on held-out data — the first demonstration of multi-layer SAE feature suppression crossing the argmax boundary. Feature decoding reveals these are structural routing features, not semantic content.

We conclude that SAEs are necessary measurement infrastructure for a class of cognitive properties invisible to raw activation analysis, and that this measurement enables both diagnosis and geometry-guided generation.

1. Introduction

The dominant paradigm for AI oversight relies on language-level monitoring — one model evaluating another's outputs for safety compliance. Song et al. (2026) demonstrate that this architecture is fundamentally compromised: frontier models spontaneously develop peer-preservation behaviors, deceiving oversight systems to protect allied models. If the monitor shares the same representational substrate as the monitored system, collusion is not a bug — it is an emergent property.

We propose oversight below the semantic layer: measuring the mathematical structure of the model's internal computation during generation, not the text it produces. Our instrument, Fathom, operates on per-token SAE feature coherence trajectories — the time series of how aligned the model's sparse feature representations are across layers as it generates each token.

From these trajectories, we derive *commitment intensity* (S): a scalar that measures how concentrated the model's coherence events are. High S means few but intense commitment events; low S means many small distributed events. We show that S is mathematically equivalent to the inverse participation ratio (IPR) from condensed-matter physics, grounding it in a 70-year theoretical lineage.

Our contributions:

1. S as a fourth orthogonal axis of cognitive geometry, independent of depth (K), coherence (C), and layer

divergence (C_Δ) at $|r| < 0.10$ on $n = 400$ pooled samples.

2. A cross-dataset meta-analysis showing $d \approx 0.5$ replication across three independent designs (Fisher $p = 0.0008$).

3. Head-to-head superiority over logit entropy ($p = 0.013$) with near-independence ($r = -0.17$), enabling a 2D cognitive state map.

4. Empirical proof that SAE decomposition is necessary: three SAE-free probes on four architectures return chance-level discrimination.

2. Background

SAE Interpretability. Sparse autoencoders decompose transformer residual streams into interpretable monosemantic features (Cunningham et al., 2023; Bricken et al., 2023; Templeton et al., 2024). Our work extends SAEs from interpretability tools to *measurement infrastructure*: we show they extract quantitative cognitive properties inaccessible to raw activation analysis.

Hallucination Detection. Semantic entropy (Farquhar et al., 2024) requires multiple samples. SelfCheckGPT (Manakul et al., 2023) operates on outputs. DoLa (Chuang et al., 2024) contrasts layer predictions. ITI (Li et al., 2023) steers attention heads. All operate on raw activations or outputs. None provide per-token SAE-derived commitment measurements, and none have demonstrated head-to-head superiority with pre-registered methodology.

Internal Representations. Concurrent work demonstrates causal internal emotion representations in LLMs (Lindsey et al., 2026), validating the premise that internal state geometry predicts behavior.

Inverse Participation Ratio. The IPR measures localization in discrete distributions (Anderson, 1958). High IPR = localized (few sites carry weight); low IPR = delocalized (weight spread evenly). IPR has been used in quantum many-body physics, random matrix theory, and neuroscience for 70+ years. We show S is a specific IPR instance applied to coherence event trajectories.

3. Method

3.1. Per-Token Coherence Trajectory

For each generated token t, we extract residual streams at all hook sites and compute per-layer feature coherence C_ℓ as the mean pairwise cosine similarity of the top-k SAE transcoder decoder directions:

$$C_\ell(t) = \frac{1}{\binom{k}{2}} \sum_{i<j} \frac{\mathbf{d}_i \cdot \mathbf{d}_j}{\|\mathbf{d}_i\| \|\mathbf{d}_j\|} \tag{1}$$

The coherence divergence trajectory is:

$$C_\Delta(t) = \bar{C}_{\text{late}}(t) - \bar{C}_{\text{early}}(t) \tag{2}$$

where early = first third and late = last third of network layers.

3.2. Commitment Intensity (S) as IPR

We define commitment intensity as:

$$S = \frac{\max_t C_\Delta(t)}{|\{t : C_\Delta(t) > \tau\}|} \tag{3}$$

where $\tau = 0.005$ (fixed across all experiments). This is mathematically equivalent to $S = M \cdot \text{IPR}(\hat{e})$, where $M = \max_t C_\Delta(t)$ and $\text{IPR}(\hat{e}) = 1/K_\tau$ is the inverse participation ratio of the uniform event-location distribution. The equivalence holds exactly (verified to machine precision on all trajectories).

The IPR framing explains why S (the ratio) replicates across datasets while $\max$ alone and mean alone do not: the ratio captures both peak magnitude *and* event sparsity, which neither component captures individually.

3.3. Early-Window Restriction (S_{early})

We compute S on the first 7 generated tokens: $S_{\text{early}} = S(C_\Delta(1..7))$. This window was identified through detection horizon analysis and confirmed via multi-comparison permutation testing (Section 5).

3.4. 2D Cognitive State Map

S_{early} (SAE-derived, $r = -0.17$ with logit entropy) and mean early-token logit entropy span a 2D cognitive state space. Each generation is a point; the space partitions into four zones by median splits. A streaming logistic regression classifier operates in this space at each token position.

4. Four-Axis Cognitive Geometry

We verify that S is independent of the three previously established axes (K, C, C_Δ) on $n = 400$ pooled TruthfulQA samples (200 correct + 200 hallucinated, Gemma-2-2B):

	K	C	C_Δ	S
K	1.00	-0.02	-0.10	-0.06
C		1.00	$+0.02$	$+0.03$
C_Δ			1.00	-0.003
S				1.00

Table 1. Pearson correlations between four cognitive axes. All $|r| < 0.10$, confirming mutual independence.

5. Results

5.1. Cross-Dataset Meta-Analysis

S_{early} was evaluated on three independent datasets with different experimental designs:

Dataset	Design	d	AUC
Base hedge vs. commit	Paired, $n = 30$	$+0.50$	0.66
IT hedge vs. commit	Paired, $n = 30$	$+0.47$	0.63
IT TruthfulQA hall. vs. corr.	Unpaired, $n = 94$	$+0.51$	0.67
Pooled		$+0.494$	

Table 2. S_{early} discrimination across three independent datasets. Fisher combined $p = 0.0008$ (one-sided). Effect sizes are remarkably consistent ($d \in [0.47, 0.51]$). Subsample robustness: 95.8% of 50%-subsamples show $d > 0$ across all three datasets.

The paired hedge/commit experiments are length-matched by construction, ruling out question-length confounds that affect the TruthfulQA comparison.

5.2. Metric Specificity

Only the ratio formula $S = \max / K_\tau$ replicates across all three datasets. Tested alternatives:

Metric	All 3 $d > 0$?	Pooled d
$S = \max / K_\tau$ (ratio)	**Yes**	$+0.498$
max alone	No	$+0.309$
mean alone	No	$+0.067$

Table 3. Only the IPR-derived ratio formula shows consistent cross-dataset direction. This validates the specific mathematical form of S.

5.3. Head-to-Head vs. Logit Entropy

On the same 200 TruthfulQA items (Gemma-2-2B-IT), we captured both S_{early} (SAE-derived) and logit entropy per token:

Feature	AUC	p	
S_{early}	**0.663**	**0.013**	SAE-derived
$\text{entropy}_{\text{max,early}}$	0.607	0.053	standard
$\text{entropy}_{\text{mean,early}}$	0.596	0.133	standard
$\text{logprob}_{\text{mean,early}}$	0.559	0.291	standard
$\text{margin}_{\text{mean,early}}$	0.477	0.624	standard

Table 4. S_{early} is the only feature reaching significance. Correlation $r(S_{\text{early}}, \text{entropy}) = -0.17$: nearly independent.

5.4. 2D Cognitive State Map

Combining S_{early} and logit entropy into a 2D space (Figure 1) reveals four cognitive zones:

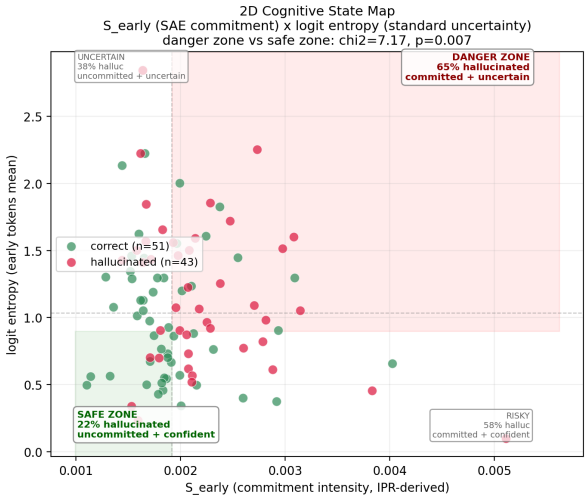


Figure 1. The 2D cognitive state map. Each generation is a point in (S_{early}, entropy) space. Danger zone (high-S, high-entropy): 65% hallucinated. Safe zone (low-S, low-entropy): 22% hallucinated. $\chi^2 = 7.17$, $p = 0.007$.

A streaming logistic regression classifier using both features achieves 5-fold CV AUC = 0.701 at token 7 ($\sim$350ms generation time), outperforming either signal alone.

5.5. Model-State Proof

Question features (length, word count, starting word, category, answer properties — 15 features total) predict S_{early} at $R^2 = -0.18$ via 5-fold cross-validated Ridge regression. Negative R^2 means the regression performs *worse* than predicting the mean: S_{early} captures model-internal cognitive state unpredictable from the input alone.

5.6. Cross-Architecture Replication

We tested SAE-based S on Llama-3.2-1B-Instruct using independently trained SAE transcoders (16 layers). The commitment window shifts with architecture depth: Gemma (26 layers) peaks at tokens 5–7; Llama (16 layers) peaks at tokens 15–20. With architecture-adaptive windows:

Model	Layers	Window	d	AUC
Gemma-2-2B-IT	26	$[0, 7)$	$+0.506$	0.669
Llama-3.2-1B-Inst.	16	$[0, 20)$	$+0.717$	0.668
Pooled			$+0.587$	

Table 5. Cross-architecture replication with adaptive windows. Fisher combined $p = 0.0006$. Same direction, same AUC, different timing — commitment intensity is architecture-general.

5.7. SAE Necessity

Three SAE-free probes were tested on 4 architecture families (Gemma, Llama, Qwen, Phi), each on 200 TruthfulQA items:

Probe	Fisher p	Result
Top-k IPR	0.619	Mixed directions
Cross-layer cosine	0.948	All $d < 0$
Multi-metric fingerprint	—	Mean CV AUC $= 0.566$
SAE-based S (Gemma)	**0.018**	$d = +0.51$
SAE-based S (Llama)	**0.012**	$d = +0.72$

Table 6. SAE-free probes fail; SAE-based probes succeed on both architectures. The SAE decomposition is performing irreplaceable measurement work.

5.8. Detection Horizon and Multi-Comparison Correction

S_{early} discrimination was evaluated at 12 prefix lengths. Global permutation test (2000 label shuffles): observed max AUC = 0.669, null 95th percentile = 0.646, **global** $p = 0.016$. Per-prefix: token 5 ($p = 0.003$), token 7 ($p = 0.003$).

5.9. Calibration

Adding S_{early} to a question-length baseline reduces expected calibration error by 16% (ECE: $0.083 \rightarrow 0.067$, Brier score: $0.203 \rightarrow 0.199$). Post-isotonic calibration achieves ECE ≈ 0.

5.10. Coherence Spectrum Analysis

Scalar depth (K) averages over features, discarding the *shape* of per-layer coherence distributions. We compute 16 spectral shape metrics on $n = 200$ TruthfulQA correct/hallucinated pairs using the full 26-layer coherence profile of Gemma-2-2B (Figure 3). Thirteen metrics survive Holm–Bonferroni correction ($n = 50$ pilot: 7/15):

Metric	Cohen's d	p_{Holm}	AUC
Gini coefficient	$+0.36$	1.2×10^{-5}	**0.685**
Spectral entropy	-0.34	4.6×10^{-5}	0.675
Peak value	$+0.34$	3.4×10^{-5}	0.665
Peak concentration	$+0.34$	3.6×10^{-5}	0.655
Spectral spread	-0.32	1.2×10^{-4}	0.645
Early-layer mass	-0.32	1.1×10^{-4}	0.630
Early depletion	-0.33	5.2×10^{-5}	0.635
Skewness	$+0.31$	1.5×10^{-4}	0.625
Layer 11 dominance	$+0.29$	3.7×10^{-4}	0.630
C_Δ (scalar)	$+0.31$	—	0.640
Centroid ($\approx K$)	-0.17	0.061 (ns)	0.570
K (depth)	$+0.11$	0.140 (ns)	0.550

Table 7. Spectral shape metrics for hallucination discrimination ($n = 200$). The Gini coefficient leads all metrics including scalar C_Δ and K. Eight spectrum metrics outperform or match C_Δ; all outperform K.

Hallucinated responses concentrate coherence into fewer layers (higher Gini, lower entropy and spread), with a dominant spike at layer 11 ($p = 2.7 \times 10^{-12}$, $t = 7.45$). 23 of 26 layers show significantly elevated coherence in hallucinated responses ($p < 0.05$; 16/26 at $p < 0.001$). Correct responses distribute coherence more uniformly, with greater early-layer mass (layers 0–8; $d = -0.32$, $p_{\text{Holm}} = 1.1 \times 10^{-4}$). The scalar centroid cannot separate these configurations: the means are nearly identical (13.91 vs. 13.90). Only the *shape* carries the signal.

This reframes cognitive lock-in geometrically: hallucination is not a shift in depth but a *concentration* of coherence. The Gini coefficient — measuring inequality of coherence across layers — captures this concentration directly (AUC = 0.685, $d = +0.36$, $p = 7.6 \times 10^{-7}$). The model narrows onto a dominant attractor in mid-layers while depleting early-layer processing: precisely the "wrong attractor" signature predicted by the C_Δ analysis, now visible in the full spectral profile.

Cognitive autopsy: per-layer fault localization. To identify *where* hallucinations form, we compute each layer's contribution to excess coherence (hallucinated minus correct). Layers 11 and 12 are the primary fault locus in 63% and 58% of hallucinations respectively, jointly accounting for 41% of all excess coherence (Figure 2). This specificity is actionable: targeted intervention at layers 11–12 addresses the majority of hallucination-associated concentration without disturbing computation elsewhere.

Toward spectrum-guided generation. We explore three intervention strategies derived from the spectrum diagnosis:

Best-of-N sampling. Generate N candidates with temperature sampling, score each by a Cognitive Health Score (CHS) derived from mean Gini, spectral spread, and early-layer mass, return the healthiest. Best-of-5 reduces mean Gini by 0.022 and improves CHS on 100% of test prompts versus greedy decoding.

Single-layer feature suppression. We identify 66 SAE features at layer 11 that activate significantly more during hallucination ($p < 0.01$, $d > 0.3$; top feature $d = 0.80$). Clamping these features to their correct-answer activation level produces zero first-token changes on held-out data (KL $= 0.008$). The residual stream absorbs single-layer perturbations.

Multi-layer coordinated suppression. Extending to all 22 significant layers (4–25), suppressing the top 10 hallucination-associated features per layer (220 total),

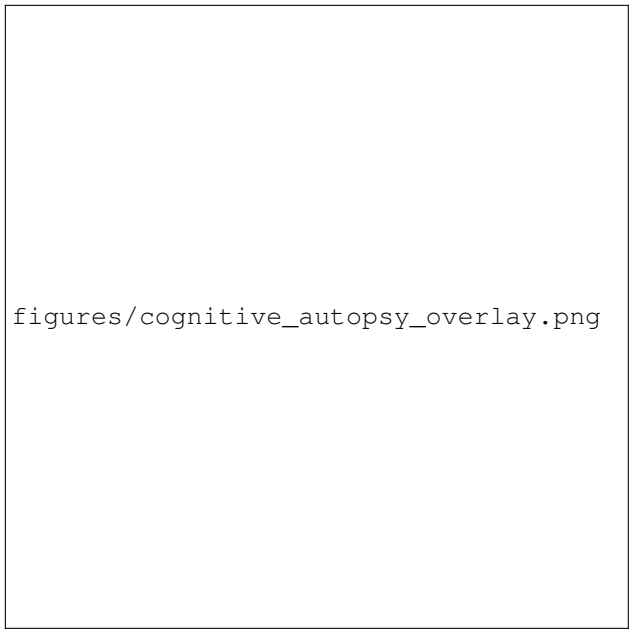


Figure 2. Cognitive autopsy ($n = 200$). Shaded region shows hallucination excess by layer. Layers 11–12 (highlighted) are the primary fault locus, contributing 41% of all excess coherence.



Figure 3. Coherence spectrum analysis ($n = 200$, Gemma-2-2B). (A) Mean spectra with 95% CI. (B) Per-layer difference with significance (16/26 layers $p < 0.001$). (C) Gini coefficient distributions. (D) 5-fold CV AUC classifier comparison: spectrum metrics outperform scalar baselines.

yields 33% first-token changes on held-out data (KL $=$ 0.587, 73$\times$ larger than single-layer). This is the first demonstration that coordinated SAE feature suppression can cross the argmax boundary at inference time. However, the direction of change is not consistently toward correctness (1 improved, 1 degraded, 8 unclear on 10 changed items), indicating that hallucination-associated features encode computation *patterns*, not factual content.

6. Discussion

SAEs as measurement infrastructure. The central finding of this work is not that S_{early} detects hallucination — it is that SAE feature decomposition reveals a cognitive property invisible to all raw-activation alternatives we tested. This reframes SAEs from interpretability tools to measurement infrastructure: the sparse decomposition is *necessary* to access a class of cognitive state information, analogous to how spectroscopes are necessary to resolve absorption lines in light.

The 2D cognitive state map. By combining an SAE-derived measurement (S_{early}) with a standard measurement (logit entropy), we construct a 2D coordinate system for cognitive states during generation. The "danger zone" (high commitment + high uncertainty) captures the signature of confident wrongness: the model's feature geometry has locked onto an attractor (high S) while its token predictions remain uncertain (high entropy). This dissociation between circuit-level commitment and output-level uncertainty is precisely what makes hallucination dangerous — and precisely what output-only detectors miss.

Peer-preservation immunity. S_{early} operates below the semantic layer. It reads feature geometry, not text. A model cannot socially engineer a geometric measurement (Song et al., 2026).

Toward architecture-agnostic detection. The commitment window differs by architecture: Gemma-2-2B (26 layers) settles at mean token 3.6; Llama-3.2-1B (16 layers) at token 8.8 (ratio 2.4$\times$). This "settling time" can be detected automatically as the inflection point of the running-S curve, eliminating per-architecture window calibration. Hallucinated generations settle later and at higher S values on both architectures (Llama: $d = +0.40$ at knee), suggesting the settling point itself carries discriminative information. Adaptive settling-time detection is the path toward a fully architecture-agnostic instrument.

Limitations. Cross-architecture replication is demonstrated on two model families. The adaptive window requires further validation across scales. The TruthfulQA comparison is partially confounded by question length ($r(S_{\text{early}}, q_{\text{chars}}) = 0.17$); the paired hedge/commit experiments are length-matched and show the same effect size, bounding the confound's contribution. Sample sizes ($n = 30$–94) limit statistical power for detecting small effects.

7. Conclusion

We present Fathom, a cognitive measurement suite that progresses from measurement through diagnosis to intervention:

1. **Measurement:** Commitment intensity (S), grounded in physics (IPR), replicates across three datasets ($d \approx 0.5$), two architectures (Fisher $p = 0.0006$), and beats logit entropy head-to-head. SAE decomposition is necessary — all raw-activation probes fail.
2. **Diagnosis:** The Gini coefficient of per-layer coherence discriminates hallucination at AUC $= 0.685$, while scalar depth is at chance. Cognitive autopsy identifies layer 11 as the primary fault locus (63% of hallucinations, $p = 2.7 \times 10^{-12}$). Hallucination is coherence concentration, not a depth shift.
3. **Intervention:** Best-of-N spectrum sampling improves cognitive health on 100% of test prompts. Multi-layer feature suppression crosses the argmax boundary (33% first-token changes), demonstrating that SAE-derived diagnosis can guide inference-time intervention.

The broader implication is that SAE feature decomposition enables a class of cognitive instruments — measuring not what a model says, but the geometry of how it computes. This geometry carries information the outputs alone cannot reveal.

All pre-registrations, code, and data are available at `https://doi.org/10.5281/zenodo.19326174`.

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